

Mikhail Razakov

Compiler Engineer

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Compiler engineer with hands-on LLVM and program-analysis experience. During my MCST internship, I implemented an LLVM 22 LICM pass and a conservative interprocedural global-IV prototype. Independent work continues the same direction: UniversalToolchain and a Roslyn ControlFlowGraph analyzer.

EXPERIENCE

July — August
2026

MCST — Compiler Engineering Intern

- Implemented an LLVM 22 LICM pass with loop analysis, side-effect and speculative-safety checks, and hoisting of loop-invariant instructions to a preheader.
- Developed a conservative interprocedural global-IV prototype with APInt affine evolution, transitive call effects, a separate legality phase, and an LLVM IR transform; unsupported CFG and call cases are left unchanged.

PROJECTS

LLVM Interprocedural Global IV Optimization

An LLVM 22 pass that localizes supported integer globals inside loops. It models affine evolution with APInt, tracks transitive call effects, and changes IR only after a separate legality check.

29/29 CI regression cases cover successful transforms and unsupported-IR rejection, LLVM Verifier checks, a second pass run, and before/after execution for runnable cases.

UniversalToolchain

Modular infrastructure for language/compiler pipelines: dependencies, providers, pass order, and artifact routes are resolved before execution and captured in a LanguagePlan.

Runtime executes the selected plan without a second resolver. The canonical test contract currently records 1,306 passing tests, with separate binding and interpreter/CIL parity checks.

DerefAfterNullAnalyzer

Roslyn ControlFlowGraph with forward fixed-point data flow: null states propagate along CFG edges and merge at join points.

Branches, loops, and back edges are supported; successors are reprocessed until the state stabilizes.

SKILLS

LLVM / compilers

C++23, LLVM 22, LLVM IR, optimization passes, CFG/SSA, dominators, loop analysis, LICM, APInt

Program analysis

interprocedural effects, legality analysis, Roslyn ControlFlowGraph, fixed-point data flow

Compiler infrastructure

C#/F.NET, language/IR pipelines, artifact routing, pass ordering, CMake, Linux

Codegen

SysV x86-64, liveness/interference, register allocation, phi lowering

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

LangDev'26 — Build the Language, Then Make the Abstractions Disappear

Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga.

UniversalToolchain — Baltic Science and Engineering Competition

First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain.

MEPhI Junior

Overall winner; First Degree Diploma, 96/100 for UniversalToolchain.

Moscow / remote